

Minsoo Cheong

Integrated M.S./Ph.D. Student in Computer Science and Engineering

Computing and Memory Architecture Laboratory (CMALab) | Seoul National University

Seoul, Republic of Korea | icycle0409@snu.ac.kr | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

RESEARCH INTERESTS

Efficient machine learning systems; large language model inference; diffusion language models; KV-cache optimization.

EDUCATION

Seoul National University · Seoul, Republic of Korea

Sep 2025 - Present

Integrated M.S./Ph.D. in Computer Science and Engineering

Advisor: Prof. Sungjoo Yoo | Computing and Memory Architecture Laboratory (CMALab)

Seoul National University · Seoul, Republic of Korea

Mar 2019 - Aug 2025

B.S. in Computer Science and Engineering, *Cum Laude*

PUBLICATIONS

EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models [\[Paper\]](#) [\[Code\]](#)

Minsoo Cheong, Donghyun Son, Woosang Lim, and Sungjoo Yoo · [\[NeurIPS 2026\]](#)

Tailoring the Quantization Space for 1-Bit KV Cache Compression [\[Paper\]](#)

Minsoo Cheong*, Donghyun Son*, and Sungjoo Yoo · *Preprint*. *Equal contribution.

KV-Rescue: Recovering Reasoning Language Model KV Eviction Loss via Stepwise Interleaving [\[Paper\]](#)

Minsoo Cheong, Woosang Lim, Vincent-Daniel Yun, and Sungjoo Yoo · *Preprint*.

Output-Aware Rotation for INT2 KV-Cache Quantization [\[Paper\]](#) [\[Code\]](#)

Vincent-Daniel Yun, Woosang Lim, **Minsoo Cheong**, Sunwoo Lee, Murali Annavaram, Sai Praneeth Karimireddy, and Sungjoo Yoo · *Preprint*.

PRESENTATIONS AND ACADEMIC SERVICE

EntropyCache. Poster presentations at NeurIPS 2026 and the ICML 2026 AdaptFM Workshop; talk at the Samsung MX Research Project Exchange Seminar, Jul 2026.

Reviewer, NeurIPS 2026 Workshop on On-Device Intelligence: Foundation Models under Real-World Constraints (ODI).

INDUSTRY EXPERIENCE

NAVER Corporation · Research Engineer Intern, LLM Solution Team

Jun 2024 - Aug 2024

Built a Kubeflow pipeline for automated LLM inference benchmarking across hardware, batching, sequence length, quantization, and speculative decoding; implemented llama.cpp support for an internal language model.

Mathpresso · Backend Software Engineer, Content Platform Team

Jun 2022 - Jun 2023

Built Kotlin/Go content-production and distribution systems and gRPC APIs; contributed to three release cycles of a digital school-exam product that achieved a 60-fold increase in sales.

Bucketplace · Backend Software Engineer, Commerce Service Team

May 2021 - Jun 2022

Developed commerce backend services using Kotlin and Spring Boot, including bulk product management and microservice data aggregation.

OPEN SOURCE & SELECTED PROJECTS

llama.cpp

Collaborator | Mar 2024 - Present

Contributions to stochastic speculative decoding, the retrieval example, and EXAONE model support in the widely used C/C++ LLM inference runtime. [\[Contributions\]](#)

krotoDC

Open-source project

A protoc plugin that generates Kotlin data classes and coroutine-friendly gRPC services and stubs, making Protocol Buffers and gRPC more idiomatic in Kotlin; distributed through Maven Central with 5,000+ monthly downloads. [\[GitHub\]](#)

TECHNICAL SKILLS

Machine Learning & Systems: PyTorch, CUDA, Hugging Face Transformers, LLM inference, KV caching, model efficiency

Languages & Infrastructure: Python, C++, Kotlin, Java; Git, Protocol Buffers, gRPC, Spring Boot, Project Reactor, Kotlin Coroutines